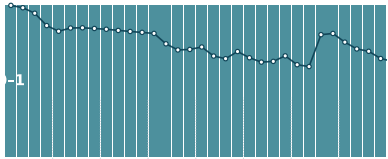
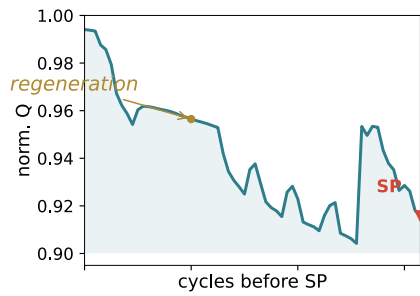
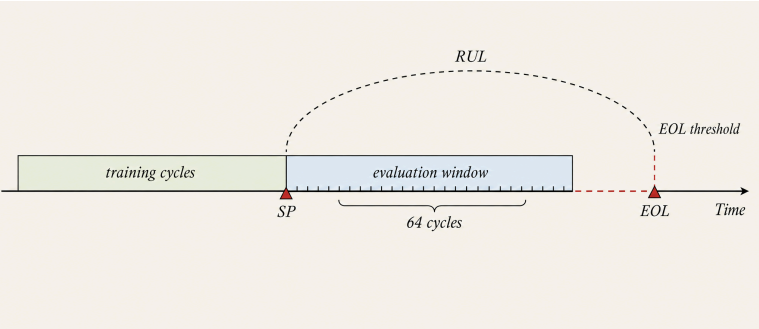
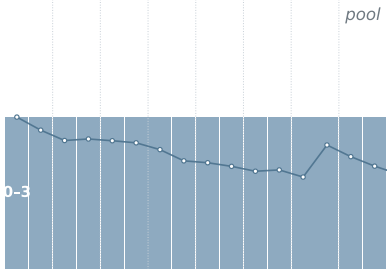


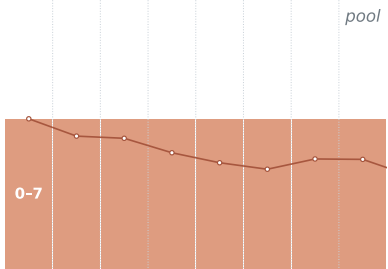
# FROM WINDOW TO TOKENS



patch 2 → 32 tokens · stride 2

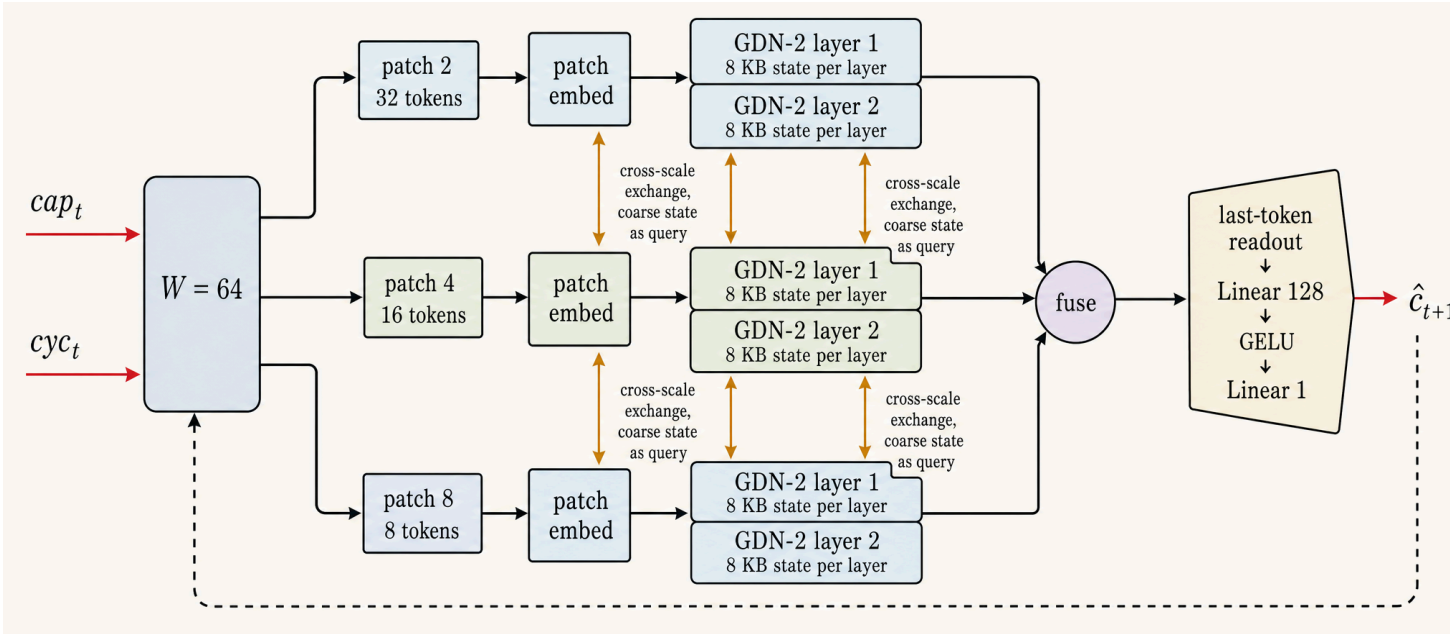


patch 4 → 16 tokens · stride 4

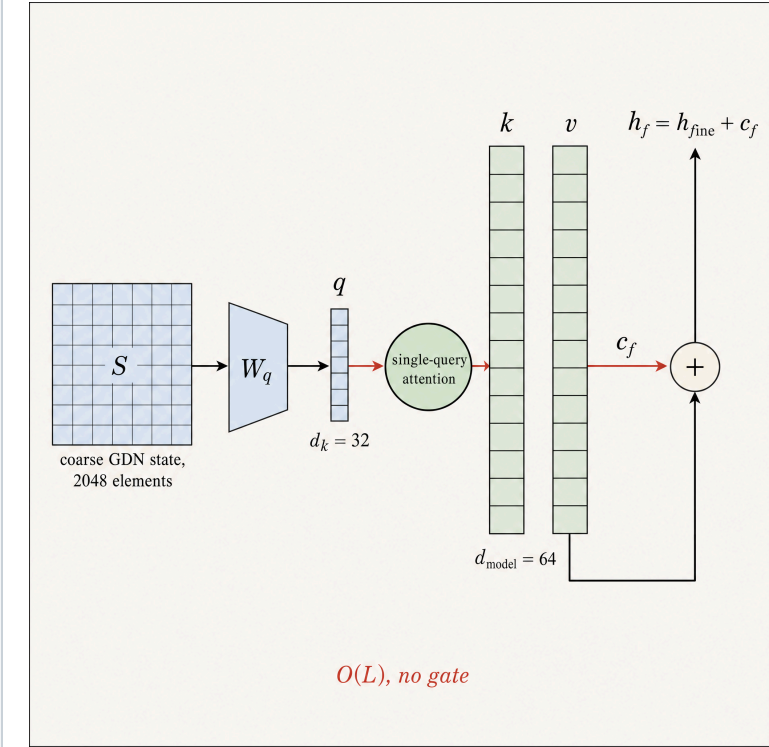


patch 8 → 8 tokens · stride 8

# MULTI-SCALE GDN-2 ENCODER · THREE BRANCHES · TWO LAYERS EACH



# STAGE-QUERY CROSS-SCALE EXCHANGE · PER LAYER



coarse state  $S \rightarrow$  single query  $\rightarrow$  sweeps fine  $k/v \cdot O(L) \cdot$  additive

$$c_f = \text{Attn}(q, K, V) \cdot h_f \leftarrow h_f + c_f$$

